Attenuation Window

Assume $0 < b \le w/2$, $x \in [0, w]$, and $y \in [0, h]$. Let the distance to the nearest horizontal edge be

$$d(x) = \min(x,\, w - x),$$

and define the smoothstep ramp

$$S(u) = 3u^2 - 2u^3, \qquad u \in [0, 1].$$

Then the attenuation window is

$$A(x) = \begin{cases} 0, & d(x) \le b, \\ S\left(\dfrac{d(x) - b}{b}\right), & b < d(x) < 2b, \\ 1, & d(x) \ge 2b, \end{cases}$$

and the desired function is

$$\boxed{f(x, y, b, w, h) = y\, A(x)}$$

so that f is zero on $0 \le x \le b$ and $w - b \le x \le w$, rises smoothly from 0 to y on $b < x < 2b$ and $w - 2b < x < w - b$, and equals y in the middle region $2b \le x \le w - 2b$.

Equivalently, in compact form,

$$f(x, y, b, w, h) = y\left[3u^2 - 2u^3\right], \qquad u = \operatorname{clamp}\left(\frac{\min(x, w - x) - b}{b}, 0, 1\right),$$

where $\operatorname{clamp}(t, 0, 1) = \max(0, \min(t, 1))$.

Here h only fixes the vertical range $y \in [0, h]$; it does not affect the horizontal attenuation. For a linear rather than smooth ramp, use $S(u) = u$.